Honest on Average, Not in Every Situation: Situation-Aware Prediction Intervals for 60-Minute CGM Glucose Forecasts

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Abstract

Forecasts of glucose 60 minutes ahead from continuous glucose monitors (CGM) can warn people with type 1 diabetes before glucose becomes dangerously low or high. A forecast is more useful when it comes with a prediction interval, or band, that states how far it can be trusted. Bands are usually judged by their average coverage, the share of true values that fall inside them, and an average can hide failures in particular situations and for particular patients. This initial draft asks whether a nominal 80% band stays honest in each situation, defined as the current glucose level combined with its 30-minute trend. Per-patient forecasters were trained for 215 people from 11 public studies, and eight of the studies (153 patients) were held out for external validation. Pooled over all studies, a standard GARCH band covered between 62% and 91% of outcomes across the situations with enough data. A situation-aware band, which learns a separate width for each situation, kept coverage between 72% and 82% and reduced per-situation miscalibration in all eight held-out studies while producing narrower bands. These results rest on one training seed per patient and are preliminary. Analyses by patient and by day are still in progress.

Status: initial draft for the Midterm Deliverable. The data and core methods are written; the remaining methods and results are planned and marked in red.

Introduction

Type 1 diabetes requires people to manage their blood glucose themselves, every day, with insulin. Too little insulin leads to hyperglycemia, conventionally glucose above 180 mg/dL, which over years damages the eyes, kidneys, nerves and blood vessels. Too much insulin leads to hypoglycemia, glucose below 70 mg/dL, which can cause confusion, seizures and loss of consciousness and can become dangerous quickly (Battelino et al., 2019). Continuous glucose monitors (CGM) measure glucose every five minutes, and that stream of readings makes it possible to forecast where glucose is heading. A forecast 60 minutes ahead is long enough to act on: there is time to eat carbohydrates before a low, or to correct a high with insulin.

A forecast is never exact, so its usefulness depends on how much it can be trusted. A prediction interval, called a band in this paper, gives a range that should contain the true future value with a stated probability. An 80% band is honest, or calibrated, if the true value falls inside it 80% of the time. A band that is too narrow gives false confidence: a person may skip a snack because the band says glucose will stay in range, and then go low. A band that is too wide triggers needless alarms and is soon ignored. Recent glucose forecasters increasingly report bands, using quantile outputs, mixture densities, Monte Carlo dropout, evidential layers or conformal methods (e.g., Sergazinov et al., 2023).

These bands are almost always evaluated by marginal coverage, the share of true values inside the band across all forecasts pooled together. Clinical use, however, is conditional. The cost of a miss depends on the situation: a forecast made at 110 mg/dL while glucose is falling fast matters far more than one made at 200 mg/dL while glucose is steady. A band can reach 80% coverage on average while covering much less in the situations that matter most, because over-coverage elsewhere hides the shortfall. In the same way, a band can be honest for the average patient but not for a particular person, or

honest across a month but not on a given day. In the literature searches for this project, only one glucose prediction study was found that reported band coverage by glucose range, in an intensive care setting, without considering the glucose trend and without proposing a correction [CITE: ICU glucose prediction study, PMC8324237]. No published study applying conformal prediction to CGM forecasting was found, and no study of ambulatory CGM forecasting was found that checks band honesty jointly by situation, by patient and by day. This is recorded as not found in these searches, not as proof that no such study exists.

This paper asks whether a band around a 60-minute CGM forecast stays honest where it is used: in each situation, for each patient, and on each day. A situation is defined as the current glucose level combined with the direction and speed of its change over the previous 30 minutes, for example "between 70 and 120 mg/dL and rising fast." Five levels and five trends give 25 situations. This initial draft addresses the first of the three questions, consistency across situations.

The draft makes three contributions. First, it describes a large evaluation setting: per-patient forecasters for 215 people with type 1 diabetes from 11 public studies, in which three datasets were used to design the methods and the other eight studies, with 153 patients, were held out and used only to check them. Second, it compares a standard band from financial econometrics, GARCH (Bollerslev, 1986), with a situation-aware band that learns a separate width for each situation, a form of group-conditional (Mondrian) conformal prediction (Vovk et al., 2005). Third, it reports initial evidence that the situation-aware band is more consistent across situations than the standard band, including on the held-out studies. Each comparison was judged against a decision rule written down before the analysis was run (Nosek et al., 2018). Because the main analysis uses one training seed per patient, these results are preliminary.

The paper is organized as follows. Section reviews related work on glucose forecasting, uncertainty for glucose forecasts and conformal prediction. Section describes

the 11 studies and how patients and records were selected. Section describes the forecaster, the situations, the bands and the measures of consistency. Section presents the initial results on consistency across situations, and Section briefly discusses them. Analyses by patient and by day will follow in later versions.

Related Work

Glucose forecasting. Short-term glucose forecasting from CGM, insulin and carbohydrate records is an active field, and most work aims at better point accuracy. Recent frameworks such as GLIMMER change how forecasters are trained so that errors in dangerous glucose ranges count more (Khamesian et al., 2026). Public datasets, including OhioT1DM (Marling & Bunescu, 2020), AZT1D (Khamesian et al., 2025) and the MetaboNet consolidation of many public studies (Wolff et al., 2026), now make it possible to test methods across many patients and settings.

Uncertainty for glucose forecasts. A smaller body of work adds bands to glucose forecasts. Gluformer produces a predictive distribution from a transformer (Sergazinov et al., 2023). Other studies use quantile outputs, Monte Carlo dropout or evidential output layers [CITE: Tan and McBeth 2026, arXiv:2603.04955; TFT quantile study, PMC12349322]. One recent study uses uncertainty to shorten the forecast horizon when glucose varies rapidly, which adapts the horizon rather than the band [CITE: MAKE 8:145, 2026, doi:10.3390/make8060145]. Personalization of uncertainty has been approached through meta-learning, which fixes a patient's uncertainty model before deployment rather than updating it during use [CITE: Zhu et al.]. Calibration in this literature is mostly reported as marginal coverage, and a 2026 review of forecasting models for type 1 diabetes calls for calibrated bands and coverage metrics in future benchmarks [CITE: Technologies 14:399, 2026, doi:10.3390/technologies14070399].

Conformal prediction. Conformal prediction builds bands from the errors a model made on held-out calibration data and guarantees average coverage under mild assumptions (Vovk et al., 2005). Conformalized quantile regression combines it with

learned quantiles to produce bands whose width varies with the input (Romano et al., 2019). Deep ensembles offer a different route, using the disagreement between several networks as a measure of uncertainty (Lakshminarayanan et al., 2017).

Group-conditional and adaptive conformal prediction. Mondrian conformal prediction calibrates separately within each predefined group, so that coverage holds within every group and not only on average (Vovk, 2012; Vovk et al., 2005). Later work extends such guarantees to broad classes of overlapping subgroups (Gibbs et al., 2023). For time series, where errors are correlated over time, adaptive conformal inference adjusts the band online after each observed hit or miss (Gibbs & Candès, 2021), and related methods target dynamic series directly (Xu & Xie, 2021). In econometrics, interval forecasts are tested for correct coverage and for clustering of misses (Christoffersen, 1998). These methods exist in general form. What this paper adds is their use with clinically meaningful glucose situations, and evidence across 11 studies on whether the resulting bands stay honest.

[PLANNED: related work on online per-patient band adjustment and on separating real miscalibration from sampling noise in small groups.]

Data

This study uses CGM records from 11 public studies of people with type 1 diabetes, 215 patients in total (Table 1). A CGM is a small sensor worn on the body that reports interstitial glucose every few minutes. The studies have two roles. Three datasets, with 62 patients, were used to develop and tune the methods. Eight further studies, with 153 patients, were never used to design or tune any method. They were used only afterwards, to test methods already fixed on the development data, as an external validation. This separation matters because a band rule tuned and judged on the same patients can look better than it is.

Development Datasets

AZT1D (Khamesian et al., 2025) contains real-world records from 25 adults using automated insulin delivery (AID), also called closed-loop therapy, in which a pump adjusts

insulin automatically from sensor readings, with a Dexcom G6 sensor. It was downloaded from Mendeley Data under a CC BY 4.0 license.

OhioT1DM (Marling & Bunescu, 2020) contains 12 adults on sensor-augmented pump (SAP) therapy, in which the pump delivers insulin as programmed and the sensor informs the user but does not steer the pump, with a Medtronic Enlite sensor. The dataset is normally released under a data use agreement. A direct access request received no reply, so the 12 patients were taken from the MetaboNet consolidation described below.

HUPA-UCM (Alvarado, 2024) contains 25 adults on pump or multiple daily injection (MDI) therapy with a FreeStyle Libre 2 sensor, with short records of about 13 days each. It was downloaded from Mendeley Data under a CC BY 4.0 license, and the preprocessed version distributed with the dataset was used.

Held-Out External Studies

The external studies come from the public subset of MetaboNet (Wolff et al., 2026), which brings 14 public type 1 diabetes datasets into one harmonized file on a common 5-minute grid. Its public subset is released for non-commercial research. Studies hosted by the Jaeb Center for Health Research follow Jaeb terms of use, and the others are under CC BY 4.0 (Jeffries et al., 2026). Eight studies were chosen to span therapy, sensor and age: Loop, DCLP3, DCLP5, PEDAP, Flair, IOBP2, ReplaceBG and BrisT1D. The MetaboNet public subset holds only the closed-loop arms of the randomized trials DCLP3, IOBP2 and Flair.

Within each study, patients with at least 45 days of CGM readings were eligible. Eligible patients were taken in a random order, and for each one the 45-day stretch with the most readings was kept. A patient was accepted if that stretch had at least 90% of the expected readings, until 20 patients per study were reached. Only 13 BrisT1D patients met these rules, which gives 153 patients overall. Their ages range from 2 to 74 years, and 69 of the 153 are under 18. The patient counts, ages, therapy labels and sensors in Table 1 come from the metadata file written when this panel was built; the age ranges describe the

sampled patients, not the whole studies.

Table 1

Studies Used in This Paper

Study	Patients	Ages (years)	Therapy	Sensor	Role
AZT1D	25	Adults	AID	Dexcom G6	Development
OhioT1DM	12	Adults	SAP	Medtronic Enlite	Development
HUPA-UCM	25	Adults	SAP or MDI	FreeStyle Libre 2	Development
Loop	20	2–65	AID	Dexcom G6/G5 (1 Libre)	Held out
DCLP3	20	14–71	AID	Dexcom G6	Held out
DCLP5	20	8–13	AID (3 SAP)	Dexcom G6	Held out
PEDAP	20	2–5	AID	Dexcom G6	Held out
Flair	20	15–27	AID	Medtronic Guardian 3	Held out
IOBP2	20	9–74	AID	Dexcom G6	Held out
ReplaceBG	20	26–74	SAP (1 unrecorded)	Dexcom G4 (1 unrecorded)	Held out
BrisT1D	13	18–24	12 of 13 unrecorded	Dexcom G6/One, Guardian 4	Held out
Total	215				

Note. AID = automated insulin delivery (closed loop); SAP = sensor-augmented pump; MDI = multiple daily injections. Ages for held-out studies are the range among sampled patients.

Signals and Preprocessing

All records are on a 5-minute grid: AZT1D and the preprocessed HUPA-UCM files are distributed this way, and MetaboNet places every study it holds on a common 5-minute grid. Four recorded signals were used: CGM glucose, basal insulin (the background rate the pump delivers, in units per hour), bolus insulin (doses given for meals or corrections) and carbohydrate intake. Other signals, such as heart rate and steps in HUPA-UCM, were not used.

Preprocessing followed the same rules for every study. Basal insulin is logged only when the rate changes, so within each patient it was carried forward to each 5-minute step. Missing bolus and carbohydrate entries were set to zero, meaning that nothing was given or eaten. Rows without a glucose reading were dropped rather than filled with an invented

value. For the held-out studies, gaps in glucose of 30 minutes or less inside the 45-day stretch were filled by linear interpolation, and rows in longer gaps were dropped. Some held-out patients had no basal record at all (10 of the 20 PEDAP patients, all 13 BrisT1D patients and one ReplaceBG patient). Their basal input was set to zero, and this is flagged in the panel metadata.

Each patient's record was cut into forecast windows of two hours of history and a target 60 minutes ahead. The windows were split in time order for each patient: the first 64% for training, the next 16% for validation and the last 20% for testing. The validation part is used for early stopping and to calibrate every band; the test part is used only for scoring. Because the split is chronological, no later data reaches the model or the band during fitting or calibration.

Caveats

Three properties of the data shape how results are reported. First, pooled numbers can be driven by one patient: in HUPA-UCM, one patient accounts for 53% of the pooled rows. For this reason results are reported per patient and per study, not only pooled over readings. Second, about 8% of pooled HUPA-UCM readings are straight-line filler, runs that look like interpolation rather than measured values. They were not removed in the analyses reported here [CHECK: confirm the filler was never excluded]. Third, therapy labels in the held-out studies are taken as recorded by MetaboNet, and they are missing for 12 of the 13 BrisT1D patients and one ReplaceBG patient.

[PLANNED: an expanded held-out panel of up to 60 patients per study, built with the same selection rules, is being analyzed; all counts above refer to the 20-per-study panel.]

Methods

This section describes the forecaster that produces the point predictions, the grouping of forecasts into situations, the bands compared, and the way their consistency across situations is measured. Throughout, a band is *calibrated* in a group of forecasts when its *coverage*, the share of true values that fall inside it, matches the stated

probability, here 80%.

Forecaster

Each patient received a separate forecasting model, following the per-patient design of GLIMMER (Khamesian et al., 2026). The model is a convolutional-recurrent network (CNN-LSTM): three one-dimensional convolutional layers (16, 32 and 16 filters, kernel size 3, ReLU activations) extract short local patterns from the input window, and a single long short-term memory (LSTM) layer with 8 units summarizes the window into one forecast. The network is small, with 4,265 trainable parameters.

The input is the 2-hour window of 24 readings described in Section . Each time step carries six variables: the four recorded signals (CGM, basal insulin, bolus insulin and carbohydrate intake), a moving average of the last 200 CGM readings (about 17 hours), and a categorical glycemic region (below 70, 70 to 180, or above 180 mg/dL). Inputs are standardized with statistics from the training windows. The target is the CGM reading 60 minutes (12 steps) after the last input reading.

Models were trained on the training windows with the Adam optimizer (learning rate 10^{-3}, batch size 64) on mean squared error for up to 30 epochs, keeping the epoch with the lowest validation root mean squared error and stopping after 5 epochs without improvement. Each patient's model had its own random seed, so that no two patients' models share the same random initialization. The main analysis uses one seed per patient.

[PLANNED: robustness forecasters (a CNN-Transformer, a 30-minute horizon, a second seed for every development patient, and one pooled forecaster trained only on the external studies), one sentence each.]

Situations

The central idea of this paper is that forecast uncertainty depends on the *situation*: where glucose currently is and where it is heading. A situation is the combination of the current glucose level and the 30-minute trend, both known at the moment the forecast is issued.

The level g_t is the current CGM reading, placed into one of five ranges: below 70 mg/dL (hypoglycemia), 70 to 120, 120 to 180, 180 to 250, and 250 mg/dL and above, where each range includes its lower edge. The trend is the change over the previous six readings (30 minutes at 5-minute sampling), $\Delta_t = g_t - g_{t-6}$, placed into one of five categories with edges at -30, -10, 10 and 30 mg/dL: falling fast ($\Delta_t < -30$), falling slowly ($-30 \le \Delta_t < -10$), flat ($-10 \le \Delta_t < 10$), rising slowly ($10 \le \Delta_t < 30$), and rising fast ($\Delta_t \ge 30$). Crossing the five levels with the five trends gives 25 situations,

$$s_t = 5\,\ell(g_t) + k(\Delta_t), \qquad \ell, k \in \{0, \ldots, 4\}, \tag{1}$$

where ℓ indexes the level range and k the trend category. Windows issued before six readings of history were available were dropped. Some situations are common (flat glucose between 120 and 180 mg/dL) and some are rare (glucose below 70 mg/dL and rising fast), which matters for how reliably each can be calibrated.

Bands

All bands are calibrated on the validation windows and scored on the test windows at a nominal coverage of 80%, so that the miscoverage rate is $\alpha = 0.2$. Every band is clipped to the 40 to 400 mg/dL reporting range of the sensor. Let $\hat{y}_t$ denote the point forecast and $e_t = y_t - \hat{y}_t$ the forecast error (residual).

Baseline: GARCH band. The generalized autoregressive conditional heteroskedasticity model, GARCH(1,1) (Bollerslev, 1986), comes from finance, where it tracks how the size of errors changes over time. Its variance follows

$$\sigma_t^2 = \omega + a\,(e_{t-1} - \mu)^2 + b\,\sigma_{t-1}^2, \tag{2}$$

with parameters μ, ω, a, b fitted on each patient's validation residuals. Because a forecast error is only known once its target time has passed, the variance is filtered with a lag equal to the 60-minute horizon and projected 12 steps ahead. The band is the Gaussian interval

$$\hat{y}_t + \mu \pm z_{0.9}\,\sigma_t, \qquad z_{0.9} \approx 1.28. \tag{3}$$

The GARCH band is used as fitted, with no further rescaling, so its average coverage depends on how well the Gaussian assumption holds for each patient.

Split conformal band. Conformal prediction (Vovk et al., 2005) sizes the band from past errors without assuming a distribution. With n validation scores $r_i = |e_i|$ sorted in increasing order, the half-width is

$$\hat{q} = r_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{4}$$

the $\lceil (n+1)(1-\alpha) \rceil$-th smallest absolute validation error, and the band is $\hat{y}_t \pm \hat{q}$. When validation and test windows are exchangeable, this guarantees at least $1-\alpha$ coverage on average over all forecasts, but not within any particular situation. The quantile was computed either from all patients pooled or from each patient alone.

Situation-aware band. The situation-aware band applies the same conformal rule separately within each situation, a form of grouped calibration known as Mondrian conformal prediction (Vovk et al., 2005). For each situation s, the half-width $\hat{q}_s$ is the conformal quantile of the absolute validation errors of the windows in that situation, and a test forecast issued in situation s_t receives the band

$$\hat{y}_t \pm \hat{q}_{s_t}. \tag{5}$$

Validation errors are pooled across the patients of a dataset within each situation, so that rare situations still have enough data. A situation with fewer than 100 validation windows falls back to the pooled quantile $\hat{q}$. Under exchangeability within each situation, this construction gives the coverage guarantee separately in every situation that has its own quantile, rather than only on average. Whether that guarantee holds up on real CGM data, where windows are not exchangeable, is what the results test.

[PLANNED: (1) a GARCH-normalized situation band, in which absolute errors are divided by σ_t before the per-situation quantile; (2) an online per-patient scale, in which the situation half-width is multiplied by $\exp(\theta_t)$ and each patient's θ_t starts at 0 and is updated once the outcome is known, one horizon later, by $\theta_{t+1} = \theta_t + \eta\,(\text{miss}_{t-12} - \alpha)$ with

$\eta = 0.01$; (3) a cold start for new patients, with situation quantiles computed from other patients only; (4) comparison baselines: deep ensembles (Lakshminarayanan et al., 2017), conformalized quantile regression (Romano et al., 2019) with and without a per-situation correction, and adaptive conformal inference (Gibbs & Candès, 2021).]

Measuring Consistency

A band can be calibrated on average while covering too little in some situations and too much in others. This paper calls a band *consistent* when its coverage is close to the nominal 80% in every situation, and measures consistency as follows.

Coverage per situation. For situation s with n_s test windows, coverage is

$$\hat{c}_s = \frac{1}{n_s} \sum_{t:\, s_t = s} \mathbf{1}\{y_t \in [L_t, U_t]\}, \tag{6}$$

where L_t and U_t are the lower and upper ends of the band.

Calibration error across situations. The summary of consistency is the average absolute distance from the nominal 80% across situations,

$$\mathrm{CE} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} |\hat{c}_s - 0.8|\,, \tag{7}$$

reported in percentage points, where $\mathcal{S}$ contains the situations with at least 30 test windows. A value of 0 means every situation is covered exactly as claimed. A companion measure, the *dispersion*, replaces 0.8 with the band's own overall coverage $\hat{c}$; it ignores a uniform shift in coverage between the validation and test periods and measures only how unevenly the band covers different situations.

Noise floor. Even a perfectly consistent band would show some spread in $\hat{c}_s$, because each situation contains a finite number of windows. This spread is larger than an independent-sample formula suggests, because consecutive forecast windows overlap and misses cluster in time. To estimate the dispersion that sampling noise alone would produce, the situation labels of each patient's test windows are circularly shifted by a random lag of at least one day (288 windows), and the dispersion is recomputed against

the unshifted sequence of hits and misses; patients with fewer than two days of test windows are left unshifted. A circular shift keeps the time structure of both the labels and the misses but breaks the link between them, so any remaining spread is chance. The *noise floor* is the mean dispersion over repeated shifts (30 to 50 in the reported analyses), and the *excess* is the observed dispersion minus the floor. An excess near zero means the remaining unevenness of a band cannot be distinguished from sampling noise.

Uncertainty and decision rules. The patient, not the forecast window, is the unit of analysis, since one patient contributes thousands of closely related windows. Uncertainty in a metric is therefore estimated with a patient-level cluster bootstrap: patients are resampled with replacement 1,000 times, the metric is recomputed on all windows of the sampled patients, and the 5th and 95th percentiles give a 90% interval. Each comparison between bands was registered with its metric, threshold and data before it was run.

[PLANNED: further analyses, one sentence each: consistency by patient and by day; error-size predictability (how much of the size of each error can be predicted from situation, patient and richer context with gradient boosting); patient difficulty (regression of log patient error on glucose variability and on the variability of 30-minute changes); and the share of error attributable to the sensor.]

Initial Results: Consistency of the Bands Across Situations

This section reports one initial result: whether an 80% band is honest in every situation, not only on average, using the situations of Section and the calibration error of Section . Each forecaster was trained with a single random seed, so all numbers below are preliminary.

Coverage by Situation

The standard GARCH band was honest on average but not by situation. On AZT1D, its coverage ranged from 64% (70 to 120 mg/dL, rising fast) to 95% (below 70 mg/dL, falling fast). Pooled over all 11 studies, the range was 62% to 91% (Figure 1, left).

The band was too narrow when glucose rose quickly from the normal range and too wide when glucose was already low and falling, both situations in which a warning could matter.

The situation-aware band was more even. Pooled over 11 studies, its coverage stayed between 72% and 82% in every situation with enough data, and between 78% and 82% in 23 of those 24 situations (Figure 1, right). The one exception, 72%, was glucose below 70 mg/dL and rising slowly.

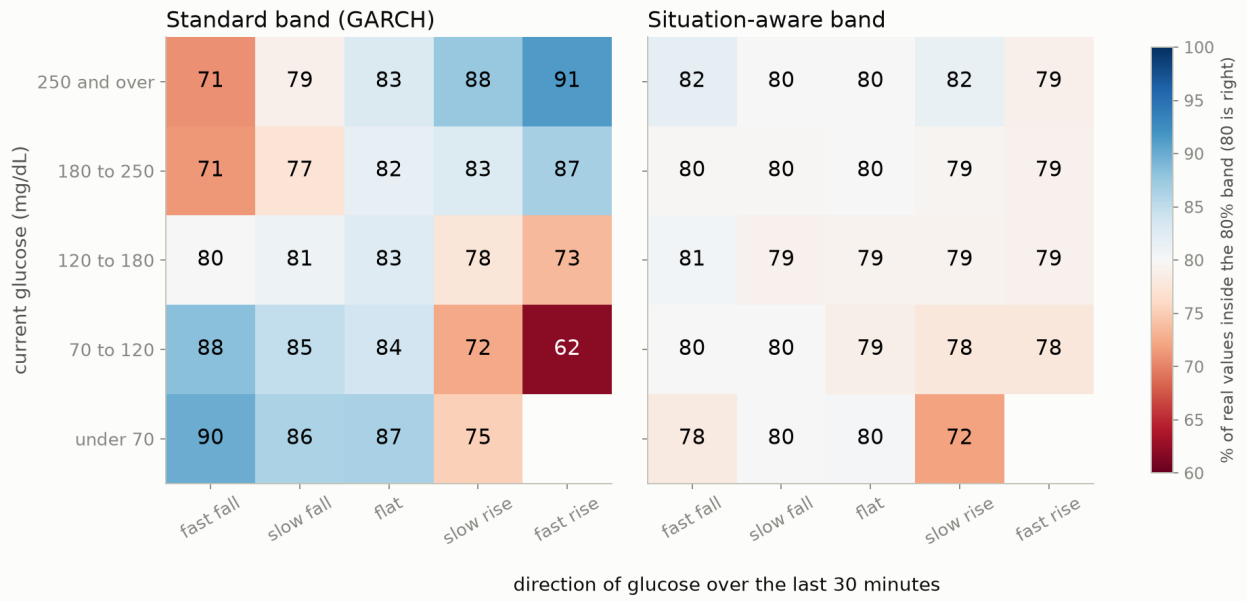


Figure 1

Share of true glucose values inside the 80% band in each situation, pooled over 11 studies. Left: the standard GARCH band (62% to 91%). Right: the situation-aware band (72% to 82%). Rows are the current glucose level and columns the 30-minute trend; a value of 80 is on target. The empty cell (below 70 mg/dL, rising fast) had fewer than 100 test windows and is not shown.

External Validation

Because the method was designed on the three development datasets, it was then tested on the eight held-out MetaboNet studies (153 patients). The rule for success was written down before the run: the situation-aware band had to show at least 20% lower

calibration error across situations than GARCH in at least six of the eight studies. The rule was met in all eight studies (Figure 2). The situation-aware band was 2.3 to 4.2 percentage points from 80% on average, against 5.5 to 6.8 points for GARCH, a reduction of between 29% and 67% depending on the study. Overall coverage stayed close to the target (78.5% to 81.2%), and the bands were narrower than GARCH in every study, by about 8 to 12 mg/dL in average full width (84 to 99 mg/dL against 94 to 108 mg/dL).

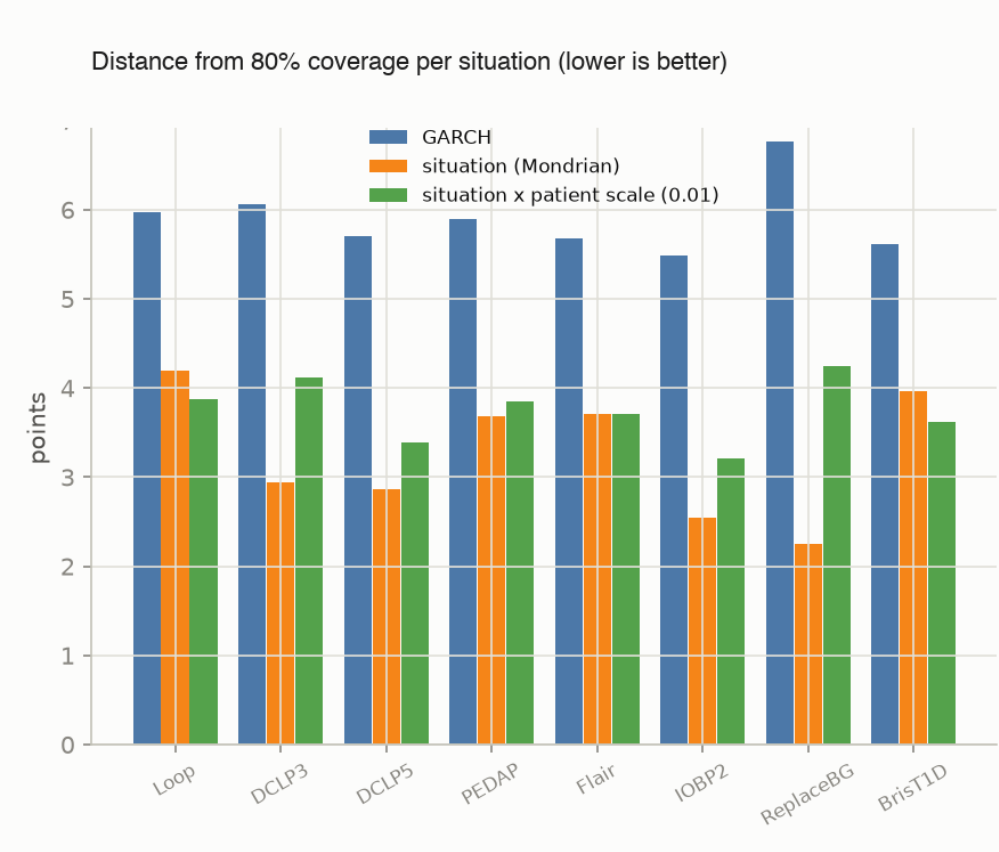


Figure 2

External validation on eight held-out MetaboNet studies (13 to 20 patients each): average distance from 80% coverage across situations, in percentage points, for GARCH (blue), the situation-aware band (orange) and the situation-aware band with a per-patient adjustment (green, a planned result). Lower is better.

Planned Results

[PLANNED: (1) the noise floor, which separates real miscalibration from sampling noise in autocorrelated forecast windows; (2) consistency across patients and the online per-patient adjustment; (3) consistency across days; (4) how predictable the size of the error is; (5) why some patients are harder to forecast than others; (6) the share of error that is unavoidable given sensor noise; (7) the trade-off between consistency and catching large, dangerous errors; (8) a second-seed check and the effect of the situation grid size.]

Discussion

The initial result suggests that a band can be honest on average while being systematically too narrow or too wide in specific situations, and that sizing the band by situation removes most of this unevenness without widening it. Because the gain held on eight studies never used to design the method, it is less likely to reflect tuning to the development data. The main limitation is that a band that is honest in every situation does not necessarily catch more dangerous forecast errors: consistency is a property of honesty, not of clinical benefit. The results also come from a single training seed per patient and remain preliminary.

[PLANNED: full discussion, including clinical relevance, comparison with prior work, and limitations of the data (45-day external records, 13 to 20 patients per study).]

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